

Trend Quality Pine Screener Pro

Professional whitepaper for a Pine v6 watchlist screener upgrade

Version	Prepared	Deliverables
v2.0 Pro	April 24, 2026	Whitepaper, tutorial PDF, and Pine v6 source file

This document redesigns the original Trend Quality Pine Screener into a stricter, better documented, Pine Screener-compatible quality and momentum model. The new version preserves the request-call budget, adds trend-template quality controls, improves relative-strength math, introduces benchmark and missing-fundamental policies, and provides clearer screener diagnostics.

Primary output file: **Trend_Quality_Screener_Pro_v2.pine**

Executive Summary

The original script already had a solid architecture: one daily symbol data request, one benchmark request, three financial data requests, and plot outputs suitable for Pine Screener. The main opportunity was not to add more external requests, because Pine Screener imposes a five request.* call ceiling. The upgrade therefore focuses on higher signal quality, safer defaults, clearer diagnostics, and better user experience while staying within that limit.

Area	Baseline	Pro v2 improvement
Screener compatibility	Exactly five request.* calls.	Keeps exactly 5 request.* calls: 2 security and 3 financial.
Trend model	Fast/slow SMA trend and slopes.	Adds optional mid-SMA alignment, yearly high/low position, and explicit trend data checks.
Relative strength	Return difference vs benchmark.	Uses multiplicative relative return: symbol return multiple divided by benchmark return multiple.
Fundamentals	Piotroski, EPS growth, revenue growth; missing values pass implicitly.	Adds a visible missing-data policy: Pass missing or Fail missing.
Workflow	Breakout or pullback entry mode.	Adds Setup only and Breakout or Pullback modes, plus status-code diagnostics.
Documentation	Working script comments.	Adds organized inputs, tooltips, score model, status legend, and tutorial.

Research Base and Platform Constraints

The redesign was constrained by TradingView Pine Screener rules and Pine v6 runtime limits. The most important findings are that Pine Screener can use indicator plots and alert conditions as filters, only existing plots can be displayed as table columns, `input.symbol()` is ignored by Pine Screener, request.* calls are capped at five in scanned scripts, and the screener calculates on the last 500 chart bars [1].

Pine v6 also supports `calc_bars_count` in declarations, which controls the most recent historical bars available to the script. The upgraded script uses `calc_bars_count = 500` to align with Pine Screener behavior and to support 200-day averages plus slope, 252-day yearly position checks, and quarterly relative-strength lookbacks [2].

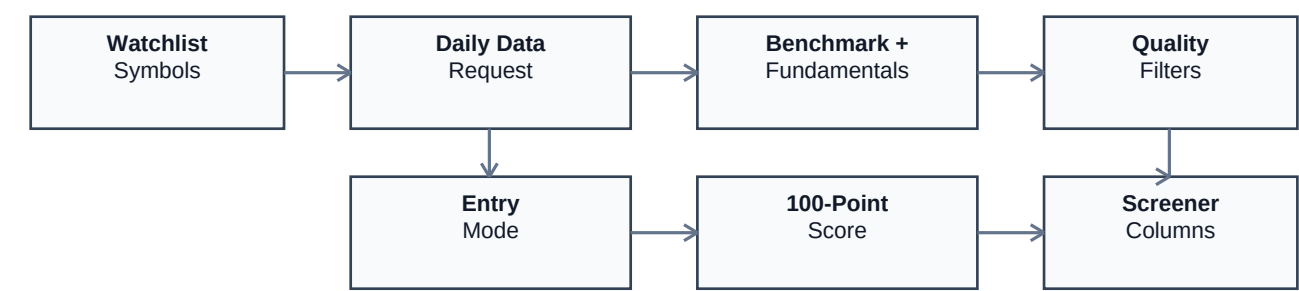
The input panel was reorganized with groups, inline controls, tooltips, and active states. These features are part of Pine input design and allow related settings to be grouped while dimming irrelevant controls when a parent option is inactive [3].

The fundamental overlay remains based on `request.financial()` metrics. TradingView documents that `request.financial(symbol, financial_id, period)` returns financial data using published financial identifiers and periods such as TTM and FQ [4]. Piotroski F-score is retained because it is a 0-9 discrete financial-strength score based on nine accounting criteria [5].

The technical model also acknowledges the long-running literature on price momentum. This is not a guarantee of future returns, but it supports the idea that relative strength and trend persistence can be useful screening dimensions when managed with risk controls [8].

System Architecture

The model is a screening tool, not an automated trading system. It reduces a large watchlist into symbols that satisfy a selected quality, trend, liquidity, volatility, relative-strength, and entry profile.



Component	Purpose	Implementation detail
Daily symbol request	Build technical state on a consistent daily basis.	Close, volume, SMAs, ATR, breakout high, return multiple, 52-week high and low.
Benchmark request	Measure relative strength and optional market regime.	Benchmark close, lookback close, fast SMA, slow SMA.
Financial requests	Add fundamental quality overlay.	EPS 1Y growth, revenue 1Y growth, Piotroski F-score.
Setup pass	Identify candidates worth chart review.	All enabled quality filters must pass.
Trade pass	Identify candidates that also meet entry mode.	Setup pass plus breakout, pullback, either, or setup-only mode.

Model Specification

Trend quality is defined by a rising fast SMA and slow SMA, price above the fast SMA, fast SMA above slow SMA, and optionally fast SMA above mid SMA above slow SMA with the mid SMA rising. The 52-week position filter defaults to requiring price within 25 percent of the yearly high and at least 30 percent above the yearly low.

Relative strength is calculated as:

$$\text{RS vs benchmark percent} = ((\text{symbol return multiple} / \text{benchmark return multiple}) - 1) * 100$$

This is more internally consistent than subtracting benchmark return from symbol return because it compares return multiples rather than additive percentage returns.

Filter	Default	Pass condition
Trend	Fast 50, mid 150, slow 200, slope 20	Close > fast SMA; fast > slow; fast and slow rising; optional mid alignment.
52-week position	Enabled	Off 52W high <= 25 percent and above 52W low >= 30 percent.
Liquidity	\$20M avg dollar volume	Close * 50-day average volume >= minimum.
Volatility	ATR 14 <= 8 percent	ATR / close * 100 <= maximum.
Relative strength	63-day vs SPY	Relative return percent >= threshold.
Fundamentals	Enabled, missing passes	Piotroski >= 6, EPS growth >= 0, revenue growth >= 0, unless missing-data policy passes missing values.
Market filter	Disabled	If enabled: benchmark close > benchmark fast SMA and benchmark fast SMA > benchmark slow SMA.
Entry	Breakout	Breakout over prior high with relative volume, pullback near fast SMA, either, or setup-only.

Scoring Model

The score remains a 0-100 composite for ranking. It is intentionally simple and transparent: a candidate earns points for each enabled category that passes.

Category	Weight	Rationale
Trend	25	Core requirement for a trend-quality screener.
Relative strength	20	Prioritizes outperformers versus the selected benchmark.
Fundamentals	15	Adds quality and growth overlay without replacing chart review.
Liquidity	10	Reduces thinly traded candidates.
Volatility	10	Controls for excessive day-to-day movement.
52-week position	10	Rewards stocks that are near highs and far above lows.
Benchmark market filter	5	Optional market-regime confirmation.
Entry trigger	5	Separates watchlist candidates from immediate trigger candidates.

Implementation Quality Improvements

Improvement	Why it matters
calc_bars_count increased from 320 to 500	Reduces false NA outcomes when yearly high/low and 200-day SMA plus slope lookback are used.
Mid-SMA alignment added inside the existing daily request	Adds a stricter trend-template layer without increasing request.* calls.
52-week position added inside the existing daily request	Filters extended downtrends and weak rebounds.
Benchmark request expanded from one value to four values	Enables relative-strength math and optional market filter while remaining one request.
Missing fundamental policy exposed	Prevents hidden pass-through behavior from accidentally admitting symbols with unavailable fundamentals.
Status code plot added	Makes failed filters diagnosable inside Pine Screener.
Input groups, inline layout, active settings, and tooltips added	Improves usability and reduces configuration errors.

Pine Screener Output Specification

The script publishes 22 plot columns and 2 alert conditions, which is safely below the 64 plot-count limit described in TradingView Pine limitations [6]. The first ten plots are the priority screener columns.

Column	Meaning	Suggested filter
Setup Pass	All setup filters pass.	= 1
Trade Pass	Setup pass plus entry trigger.	= 1 for actionable list
TQ Score	0-100 composite rank.	Sort descending; consider >= 80
RS vs Benchmark %	Relative return advantage over benchmark.	>= 0, or stricter >= 5

% From Fast SMA	Extension above/below fast SMA.	Use to avoid overextension
ATR %	ATR relative to price.	<= configured risk threshold
Avg \$ Volume	Average dollar liquidity.	>= required trading liquidity
Rel Volume	Current volume / average volume.	>= 1.5 for breakout mode
Piotroski F	0-9 financial-strength score.	>= 6, or stricter >= 7
EPS 1Y Growth %	One-year EPS growth.	>= configured threshold

Status code legend: 0 = pass, 1 = insufficient price data, 2 = trend, 3 = liquidity, 4 = volatility, 5 = relative strength or benchmark, 6 = 52-week position, 7 = fundamentals, 8 = market filter, 9 = entry trigger.

Validation Checklist

- The script contains exactly 5 request.* calls: 2 request.security() calls and 3 request.financial() calls.
- The total tuple element load is approximately 19, well below Pine Script combined request tuple limits.
- The script contains 22 plot() calls and 2 alertcondition() calls, comfortably below the maximum plot-count ceiling.
- All request.security() calls use lookahead = barmerge.lookahead_off to avoid forward-looking values.
- The benchmark input is input.string(), not input.symbol(), because Pine Screener ignores input.symbol() selections.
- The screener is an indicator, not a strategy. It does not place orders or produce backtest performance statistics.

Limitations and Risk Notes

A screener can improve consistency, but it cannot remove market risk. Financial data can be missing or delayed for some symbols. Relative-strength leadership can reverse. Breakouts can fail. High scores should be treated as a prioritization signal for chart review, not as standalone trade advice.

The script should be tested on the exact market, asset type, benchmark, timeframe, and liquidity universe that the user intends to scan. Recommended practice is to compare strict and loose presets, review false positives, and adjust thresholds before using alerts.

Source Excerpt

The complete source is delivered separately as Trend_Quality_Screener_Pro_v2.pine. The excerpt below shows the request architecture and score model.

```
indicator("Trend Quality Pine Screener Pro", shorttitle = "TQ Screener Pro", overlay =
    false, calc_bars_count = 500, precision = 2)
[dClose, dPrevClose, dVolume, smaFast, smaMid, smaSlow, smaFastPast, smaMidPast,
    smaSlowPast, atrValue, avgVolume, prevBreakHigh, symbolRetMultiple, high52, low52] =
    request.security(
[benchClose, benchPastClose, benchSmaFast, benchSmaSlow] = request.security(
epsGrowth = request.financial(
revGrowth = request.financial(
piotroski = request.financial(
score += trend0k      ? 25.0 : 0.0
score += rs0k         ? 20.0 : 0.0
score += liquidity0k  ? 10.0 : 0.0
score += volatility0k ? 10.0 : 0.0
score += position0k   ? 10.0 : 0.0
```

```
score += fundamentals0k ? 15.0 : 0.0
score += market0k      ? 5.0  : 0.0
score += entry0k        ? 5.0  : 0.0
alertcondition(
alertcondition(
```

References

- [1] **TradingView Help Center**. Pine Screener: key features and requirements.
<https://www.tradingview.com/support/solutions/43000742436-tradingview-pine-screener-key-features-and-requirements/>
- [2] **TradingView Pine Script docs**. Declaration statements: calc_bars_count and dynamic_requests.
<https://www.tradingview.com/pine-script-docs/language/declaration-statements/>
- [3] **TradingView Pine Script docs**. Inputs: group, inline, active, and input organization.
<https://www.tradingview.com/pine-script-docs/concepts/inputs/>
- [4] **TradingView Help Center**. What financial data is available in Pine. <https://www.tradingview.com/support/solutions/43000564727-what-financial-data-is-available-in-pine/>
- [5] **TradingView Help Center**. Piotroski F-score. <https://www.tradingview.com/support/solutions/43000597734-piotroski-f-score/>
- [6] **TradingView Pine Script docs**. Limitations: plot counts, request calls, and tuple element limit.
<https://www.tradingview.com/pine-script-docs/writing/limitations/>
- [7] **TradingView Pine Script docs**. Style guide. <https://www.tradingview.com/pine-script-docs/writing/style-guide/>
- [8] **Jegadeesh and Titman**. Momentum: Evidence and insights 30 years later.
<https://ideas.repec.org/a/eee/pacfin/v82y2023ics0927538x23002731.html>